

What HTML is

HyperText Markup Language. A *markup* language annotates content — it labels what each piece is. It does not compute, decide, or behave.

HTML is not a programming language. No variables, no loops, no decisions. Every line is a declaration, not an instruction.

Skeleton analogy. Every page on the internet uses the same underlying structure — the same bones. What differs is the content inside them.

LANGUAGE	JOB
HTML	structure — what each thing is
CSS	presentation — how it looks
JavaScript	behavior — what it does

Tags, elements, attributes

Tag

The code itself. `<p>` and `</p>` are two tags.

Element

Tag + content. `<p>Hello</p>` is one element.

Attribute

Extra info inside the opening tag. `name="value"`.

```
<a href="https://example.com">Click here</a>
```

Diagram labels for the example code above:

- tag: `<a`
- attr name: `href`
- attr value: `"https://example.com"`
- content: `Click here`
- closing tag: `</a>`

Attributes go in the opening tag only. Never `</a href="...">`.

The four essential parts

PART	WHAT IT DOES	VISIBLE?
<code><!DOCTYPE html></code>	Tells the browser this is HTML5. Must be the very first line.	No
<code><html lang="en"></code>	The root element. Everything else lives inside it.	—
<code><head></code>	Metadata: title, character set, viewport, CSS links.	No
<code><body></code>	Every visible thing — headings, paragraphs, images, links.	Yes

The boilerplate

```

<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport"
      content="width=device-width,
      initial-scale=1.0">
    <title>Document</title>
  </head>
  <body>
    <!-- everything visible goes here -->
  </body>
</html>

```

```
<!DOCTYPE html>
```

```

<head>
<meta charset>
<meta viewport>
<title>
</head>

```

```

<body>
h1, p,
img, a, ul,
everything
visible

</body>

```

head is invisible, body is not

Emmet shortcut: type `!` then Tab in VS Code. The entire boilerplate appears instantly. Emmet encodes the four parts above — nothing more.

The head — metadata

ELEMENT	PURPOSE
<code><title></code>	Shows in the browser tab. The only head element users ever see.
<code><meta charset="UTF-8"></code>	Character encoding. Standard choice for English and most languages.
<code><meta name="viewport"></code>	Makes the page responsive on phones and tablets.
<code><link></code>	Attaches a CSS stylesheet. (Unit 2.)
<code><script></code>	Attaches a JavaScript file. (Later units.)

The body — everything visible

If it is not in the `<body>`, the user does not see it. That single rule prevents a dozen beginner bugs.

ELEMENT	PURPOSE
<code><h1></code> – <code><h6></code>	headings
<code><p></code>	paragraphs
<code></code>	images
<code><a></code>	links
<code></code> · <code></code>	lists

Header order matters. `<head>` comes before `<body>`. Swapping them produces a broken-looking page with no obvious cause.

Self-closing tags. `<img>`, `<br>`, `<hr>`, and `<meta>` have no closing tag. There is no content to wrap.

Five mistakes that break the page

MISTAKE	WHAT HAPPENS
Missing closing tag	Every <code><html></code> needs its <code></html></code> . Missing tags cascade.

Wrong order	<code><head></code> before <code><body></code> , always.
Missing DOCTYPE	Browser goes into "quirks mode" and renders like it is 1998.
Wrong file extension	<code>.html</code> , not <code>.txt</code> or <code>.htm</code> .
Typos in tag names	<code><html></code> is not a tag. Neither is <code><bdy></code> .

On the exam

HTML is not tested directly on the AP CSP exam. The exam uses College Board's pseudocode for algorithm questions. HTML appears on the *Create Performance Task* if you build a web-based project — and every real-world project uses these exact four parts.

What is next. The skeleton exists. It has no content yet. Lesson 3 fills it with text — headings, paragraphs, and emphasis — and introduces the whitespace rule that explains why your line breaks vanish.

Keep this sheet. Every later HTML lesson assumes the four parts are on it without being told where they are.